

Biology Mock 1 — Units 2 and 3

MTOH • 90 MINUTES • 80 MARKS

NAME

DATE

SCORE / 80

- Answer every question in the space provided. Show your working and your reasoning — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- The mark for each question is shown in square brackets on the right. There are 80 marks in 90 minutes.
- Where a question says explain, a feature on its own is not enough — say what it achieves. “The wall is thin” is a description; “thin, so the diffusion distance is short” is an explanation.
- Where a question asks for a sequence, write it in order and make each step the cause of the next. Naming only the first and last steps scores the marks for neither.
- Section 1 is recall, Section 2 is explanation, Section 3 is data and problems. Work in order and do not spend more than six minutes stuck on any one question.

SECTION 1 – THE MECHANICS

1. The thorax and its muscles.

- (a)** Name the muscle that forms the floor of the thorax. [1]

- (b)** State what this muscle does during inhalation. [1]

2. Complete these figures for the composition of expired air.

- (a)** Approximate percentage of oxygen. [1]

- (b)** Approximate percentage of carbon dioxide. [1]

3. Blood returning from the lungs.

(a) Name the blood vessel that carries blood from the lungs to the heart. [1]

(b) Name the chamber of the heart that this blood enters. [1]

4. Valves in the heart.

(a) Name the valve between the right atrium and the right ventricle. [1]

(b) State the function of this valve. [1]

5. Name two components of blood other than plasma, and state one function of each. [2]

6. Define each of these ecological terms.

(a) Habitat [1]

(b) Community [1]

7. A food chain in a lake is: algae → water flea → small fish → heron.

(a) Name the producer. [1]

(b) State the trophic level of the small fish, as a number and as a name. [1]

8. Decomposers.

(a) Name the two groups of organisms that act as decomposers. [1]

(b) State what decomposers return to the soil. [1]

9. Greenhouse gases.

(a) Name one greenhouse gas other than water vapour. [1]

(b) State one human activity that increases the concentration of the gas you named. [1]

10. Testing for carbon dioxide.

(a) Name the solution used, and state what you would see if carbon dioxide is present. [1]

(b) State whether inspired or expired air produces this change faster, and why. [1]

- 11.** State two differences between inspired and expired air, other than the oxygen content. [2]

Two marks, two differences. One of them is not a gas at all.

SECTION 2 – EXPLANATION

- 12.** Describe, in order, what happens in the thorax during inhalation. Begin with the muscles and end with the movement of air. [3]

Three marks, and the order is the mark scheme. Muscles change volume; volume changes pressure; pressure moves the air.

- 13.** Explain three ways in which an alveolus is adapted for rapid gas exchange. In each case state what the adaptation achieves. [3]

- 14.** Explain why the wall of the left ventricle is much thicker than the wall of the right ventricle. [3]

Start by stating what is the same about the two ventricles. That sentence is worth a mark and it stops you writing the usual wrong answer.

15. Compare an artery with a vein. Give three structural differences.

[3]

16. Blood travels more slowly through the capillaries than through any other vessel.
Explain why, and explain why this is an advantage.

[3]

17. Describe the path taken by a drop of blood from the vena cava to the aorta,
naming every chamber and vessel it passes through, in order.

[3]

18. Explain the difference between the way energy moves through an ecosystem and
the way nutrients move through it.

[3]

One of the two words in bold goes in a straight line and one goes round in a circle. Say which, and say what happens at the end of the straight line.

19. Explain why food chains rarely contain more than four or five trophic levels. [3]

20. Describe how a student would use a quadrat to estimate the number of dandelions in a field. [3]

21. Explain how cutting down a forest raises the concentration of carbon dioxide in the atmosphere. Give two separate mechanisms, then state one other consequence of deforestation. [3]

22. Describe how the greenhouse effect warms the Earth. Write it as three linked steps. [3]

23. At rest, Priya takes 14 breaths a minute and moves 480 cm^3 of air with each breath. Her heart beats 68 times a minute and pushes out 72 cm^3 of blood per beat.

(a) Calculate her ventilation rate, in dm^3 per minute. [2]

(b) Calculate her cardiac output, in dm^3 per minute. [2]

(c) During exercise both of these values rise. Explain why. [1]

Both calculations are the same shape: how many times a minute, multiplied by how much each time. Convert to dm^3 once, at the end.

24. A healthy pair of lungs contains 3.0×10^8 alveoli, each with a surface area of $2.4 \times 10^{-7} \text{ m}^2$.

(a) Calculate the total surface area available for gas exchange, in m^2 . [2]

- (b)** A patient with emphysema has 9.0×10^7 alveoli, each with a surface area of $4.0 \times 10^{-7} \text{ m}^2$. Calculate his total surface area and the percentage reduction compared with a healthy pair of lungs. [2]

- (c)** Explain why he becomes breathless climbing one flight of stairs. [1]

Multiply the numbers, then add the powers of ten. Note that his individual alveoli are larger — and his total area is still smaller.

25. A food web in a meadow contains these feeding relationships:

grass → grasshopper → shrew → owl · grass → rabbit → fox · shrew → fox

- (a)** Name the organisms at trophic level 2. [1]

- (b)** A disease kills every rabbit. State and explain what happens to (i) the grass and (ii) the fox population. [2]

- (c)** Use this web to explain why a food web is more stable than a single food chain. [2]

26. The producers in a lake capture 48 000 kJ of energy per square metre per year. About 10% of the energy at each trophic level is passed on to the next.

(a) Calculate the energy available to the primary consumers, in kJ/m²/year. [1]

(b) Calculate the energy available to the tertiary consumers, and express it as a percentage of the energy captured by the producers. [2]

(c) A fish farm grinds small fish into pellets and feeds them to larger fish, which are then sold. Explain, in energy terms, why selling the small fish for people to eat would feed more people. [2]

27. Water was sampled downstream of a pipe carrying run-off from a fertilised field. Fish need more than 5 mg/dm³ of dissolved oxygen to survive.

Distance / m	0	100	300	600	1200
Nitrate / mg dm ⁻³	48	41	26	11	4
Oxygen / mg dm ⁻³	8.8	2.6	1.4	4.9	8.1

(a) State the distance at which the dissolved oxygen is lowest. [1]

(b) Calculate the percentage decrease in nitrate concentration between 0 m and 600 m. [2]

(c) Explain, in order, how the nitrate in the water causes the fall in dissolved oxygen.

[2]

In (c) there are five links between the nitrate and the oxygen, and none of them involves anything poisonous.

End of paper. Go back to every question that asked you to explain and check that each feature you named is followed by what it achieves. That is where this paper is won and lost.